Positive-unlabeled learning with knowledge integration. Application to mineral prospectivity mapping

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Talk map: 30-month PhD program progress

1 MPM problem

- Introduce a real mineral exploration problem.
- Transform to Machine learning problem

2 General challenges

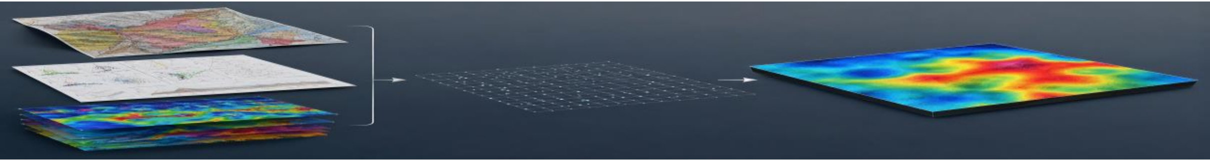
- Positive–Unlabeled learning
- Knowledge integration into machine learning.

3 Scientific progress + BRGM application

- Graph-based MPM baseline.
- Neuro-symbolic PU method.
- Publications / submissions.
- Application to BRGM datasets (Armorican dataset, Vosges dataset)

4 Next steps

- Promising approaches/ Additional experiments
- How can this thesis work be reused at BRGM



Mineral Prospectivity Mapping (MPM)

- **Input Data**: Geological data over a particular region

Geological Formation | Faults | Mineral Occurrences **(Ground-Truth Labels)**

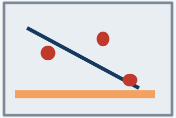
- **Goal:** Construct a Prospectivity Mineralization Map

Prospectivity Mineralization Map

MPM to Machine Learning Problem

Cell-Based Analysis (CBA) : transform MPM to Machine Learning (ML) problem

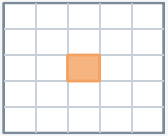
1 Study area



Geological layers
+ known occurrences

Input: maps from faults, geochemistry, geophysics

2 Cut into cells



One cell
= one small region

Example in Vosges: 2500 m × 2500 m per cell

3 Describe each cell

rock type
fault present
Cu value
magnetic value

For each cell, summarize the evidence

This creates numbers that a computer can use

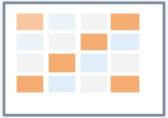
4 Add known deposits

cell	deposit?
A	yes
B	?
C	?

Known deposit
= positive

All other cells are not "negative"; they are unknown

5 Learn a map



Output:
prospectivity score

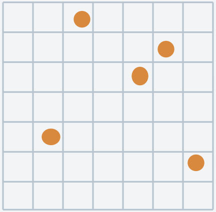
The model ranks cells from less promising to more promising

Main idea CBA turns a map into a table: each cell becomes one example, described by geological evidence and a label to show a deposit is known.

Challenge: MPM to General PU Problem

MPM viewpoint

Study area divided into pixels



Known deposits

= reliable positive pixels

All other pixels

= unknown, not true negatives (unexplored region)

Reformulate

General Positive Unlabeled (PU) learning

Observed data

P
known positives

U
unlabeled

But U is a mixture:

hidden positives + real negatives

Learning objective:

predict prospectivity score without verified negatives

Main difficulty

No pixel can be confidently labeled as "non-mineralized".

So classical supervised learning needs pseudo-negatives, but their quality is uncertain.

Challenge: generalize MPM to PU learning

Why PU problem is challenging in MPM

The general PU formulation is useful, but MPM adds geological and domain-specific constraints

Key question *How can we learn a reliable prospectivity model when positives are rare, negatives are unknown, and geological knowledge must be respected?*

1. Label uncertainty

- Known deposits are positives, but *absence of deposit* does not mean negative.
- Random pseudo-negatives may include hidden positives.

2. Spatial dependence

- *Mineral occurrences* tend to cluster spatially.
- However, *geological continuity* can create spatial bias in training and evaluation.

3. Knowledge integration

- *Expert rules from geologists* are useful but imperfect.
- Rules should guide the model without forcing wrong conclusions.

MPM data — **PU problem** — **PU + knowledge integration**

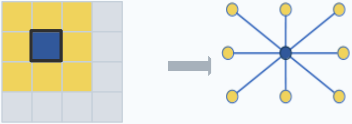
SOTA data-driven MPM: strengths and limitations

Current SOTA **Graph Neural Networks (GNNs) are a strong data-driven baseline for MPM.**

Strengths

- Represents the study area as a graph of cells or regions.
- Uses neighborhood relations between cells to captures local spatial structure.
- Shows strong predictive performance in recent MPM studies.

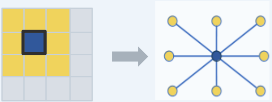
Simple view of the idea



Cells on a map   Cells as a graph

Limitation 1 — Graph construction is often too simple

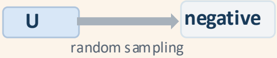
- Neighbors are often chosen only by spatial distance.
- Nearby cells may be linked even when their geology is different.



distance only   spatial neighbors

Limitation 2 — Label setting is difficult

- Only a few cells are known positives.
- Most other cells are unlabeled, not true negatives.
- Random negative sampling can introduce bias and instability.



Result: the model may depend too much on one random choice of negatives.

Current SOTA is powerful, but it can be improved in both graph construction and handling of unlabeled data.

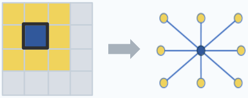
1st-year work: Improve GNN Network in MPM

Contribution 1: Novel graph construction for MPM

Graph structure

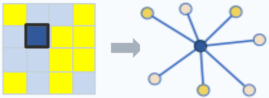
- Do not rely only on spatial distance.
- Build links using geological attributes (faults, formations, rock types, etc.) besides spatial distance
- *Goal*: create more meaningful neighbors for the GNN.

Existing graph construction



spatial　　neighbors

Proposed

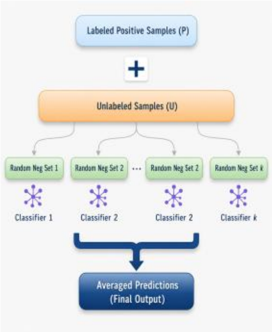


geological　　better neighbors

Contribution 2: Ensemble strategy for PU learning

PU learning

- Train k classifiers.
- Each classifier uses a different sampled negative set.
- Average the predictions at the end.
- *Goal*: reduce bias from one random negative sample.



1st work: Improve Graph Neural Network in MPM

- Evaluate the proposed methods on

Investigate Antimony Occurrences in Armorican Massif

=> Show the better performance than current models (3%-5% on almost metrics)

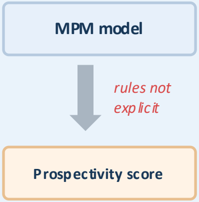
Publication:

Thi Hai Yen VU, Thi Bich Hanh DAO, Vincent Nguyen, Christel VRAIN, Hugo BREUILLARD (2025). Features Leverage in Graph Models for Mineral Prospectivity Mapping. InProceedings of the 40th ACM/SIGAPP Symposium on Applied Computing (SAC 2025).

Limitations of previous work

1. Expert knowledge is not explicit

Previous MPM models can learn from data, but they do not explicitly force predictions to follow rich expert rules.



Example rules from geology

w1: Close_crustal_fault(R) → Pathway(R)
IF close to crustal-scale faults THEN pathway favorability is high

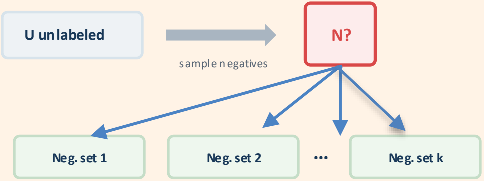
w2: Fertile(R) → Source(R)
IF fertile lithology is present THEN source favorability is high

w3: Source(R) ∧ Pathway(R) → Prospectivity(R)
IF Source AND Pathway are present THEN Prospectivity is high

These rules are useful, but previous models treat them only implicitly or after prediction.

2. PU learning: negative sampling uncertainty

In MPM, known deposits are positive, while the rest of the map is unlabeled. Sampling negatives from U is risky.



Quality of negative sampling is uncertain

Hidden positives may be sampled as negatives

Motivation: move toward a knowledge-aware PU framework that can use expert rules and reduce dependence on one sampled negative set.

Our current Proposal
NeuPSL: A NeuroSymbolic Framework

NeuPSL : logic reasoning + neural network learning

Expert rules & geology:

w1: Close_crustal_fault (R) -> Pathway (R)

*(**IF** region R is close to crustal-scale faults **THEN** pathway favorability is high)*

w2: Fertile (R) -> Source (R)

*(**IF** region R has fertile lithology present **THEN** source favorability is high)*

w3: Source (R) /\ Pathway (R) -> Prospectivity (R)

*(**IF** Source AND Pathway in region R are favorable **THEN** Prospectivity is high)*

w4: Neural (R) -> Prospectivity (R)

*(**IF** result of **data-driven** model NEURAL of region R is good **THEN** Prospectivity is high)*

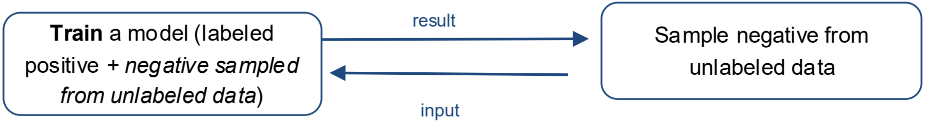
Probabilistic Soft Logic(PSL)

Find the best value Prospectivity (R), all the weights w1, w2, w3, w4 (importance/ reliability of rule)

Our solution for PU learning

Our solution: **Form an iterative loop**

1. ***Train a NeuPSL model***
2. ***Learned model : => choose negative = samples with least favorability score***
3. *From negative examples sampled => train a new model*



2nd work: Theorical PU learning for general Neuro Symbolic framework (not only for MPM)

- Formulate PU learning problem on general Neuro Symbolic (NeSy) Framework
- Construct necessary foundation for PU learning in Data with knowledge

  => propose the ***NeuralAdjust*** method
- Do experiments on some public dataset for paper and citation with simple knowledge rules

  => Create PU data setting for experiments:

  + Only some observed positive papers (e.g. topic AI)

  + All the rest = unlabeled (e.g. positive = topic AI, negative = other topics)

  => Evaluate predictions on unlabeled papers

2nd work: Theorical PU learning for general Neuro Symbolic framework (not for MPM)

Our solution outperformed State-of-the-art models on the same data setting for popular datasets (e.g. Cora, Citeseer, Pubmed, Chamelon)

Publication:

Thi Hai Yen VU, Vincent Nguyen, Christel VRAIN, Hugo BREUILLARD, Thi Bich Hanh DAO

Positive-Unlabeled Learning with Knowledge-Structured Data

European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases

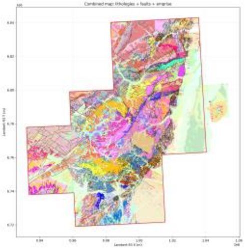
~ *Acceptance rate: 24% (conference rank A in Informatics)*

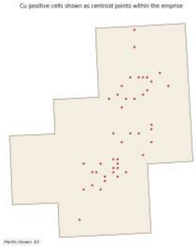
3rd work: Application on BRGM dataset & theorical study on noisy rules

- *IRM project:*

  - Study *NeuralAdjust method* (2nd-year work for PU learning on data with knowledge from geologist) on MPM problem

  - Application on Vosges dataset





3rd-year work: Application on BRGM dataset & theorical study on noisy rules

Goal: study how imperfect knowledge rules affect model predictions and robustness.

Main question — **Does the model still benefit from expert knowledge when part of the rule set is noisy?**

1. Start from clean expert rules

Clean Knowledge

Examples of knowledge rules:

w1: Fault(R) → Pathway(R)
w2: Fertile(R) → Source(R)
w3: Source(R) ∧ Pathway(R) → Prospectivity(R)

The rules represent geological assumptions that should guide the MPM prediction.

2. Inject controlled noise

Noisy Knowledge

Create several noisy rule sets:

η = 0%	η = 10%	η = 20%	η = 40%

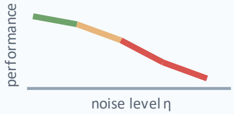
Noise examples:

- *Add wrong rules*
- *Remove useful rules*
- *Change rule confidence / weight*

3. Train and evaluate

Robustness Test

- *Train the model*
- *Compare against clean-rule baseline.*
- *Observe the performance drop.*



expected: gradual decrease

Expected outcome: *quantify the impact of noisy knowledge and identify how much rule noise the model can tolerate.*

Conclusion

MPM problem

A real mineral exploration problem

1. **Few known occurrences**
   Only a small number of deposits are observed.

2. **No verified negatives**
   Unexplored cells are unlabeled, not true negatives.

3. **Expert knowledge exists**
   Geologists know useful rules, but they are not automatically used by ML.

Objective 1 : PU learning

Learn from positives and unlabeled data because reliable negatives are unavailable.

Addresses: few occurrences + no verified negatives

Objective 2: Knowledge Integration

Use geological rules to guide learning because data are scarce and expert knowledge is valuable.

Addresses: scarce data + explicit expert reasoning

Works developed in the thesis

1. Graph-based MPM

Geological graph construction + ensemble PU

2. Neuro-symbolic PU

NeuralAdjust / NeuPSL formulation

3. BRGM applications

Armorican and Vosges datasets

4. Rule robustness

Study of noisy knowledge rules

Research perspectives

Thesis outputs
PU + expert knowledge for MPM

→ **Four possible research directions**

1. Transfer to new BRGM cases

- Other commodities or study areas
- Test transferability between geological contexts

2. More robust knowledge use

- Noisy, missing or uncertain rules
- Build and update reusable rule bases

3. Improve model reliability

- Better negative selection in PU learning
- Uncertainty and interpretability of maps

4. Prototype for decision support

- Reusable workflow or internal demonstrator
- Support geologists, not replace them

Thanks for your attention!!